

Shandeesh R P

Erode, Tamil Nadu, India | LinkedIn | GitHub | Email | Phone

SUMMARY

Motivated Computer Science student with hands-on experience in Cybersecurity, Security Operations, AI/ML, Python development, web security, IoT security systems, and software engineering. Experienced in developing security-focused applications, analyzing SOC incidents, building AI-driven solutions, integrating hardware and software systems, and working with technologies such as Python, Java, PyTorch, XGBoost, FastAPI, Next.js, Streamlit, Raspberry Pi, NFC, sensors, Burp Suite, Nmap, Bettercap, Wazuh, and MT5. Strong interest in SOC operations, threat detection, vulnerability assessment, AI-powered cybersecurity, and secure application development.

EDUCATION

Nandha Engineering College, Erode, Tamil Nadu

Bachelor of Engineering – Computer Science and Engineering (Cyber Security), Affiliated to Anna University, Chennai, Relevant Coursework: Cybersecurity & Secure Software Engineering, Data Structures, Database Management Systems, Computer Organization & Architecture, Software Engineering, Python Programming, Web Technologies, Network Security, Database Security

EXPERIENCE

Cybersecurity Intern

Mindenious Edutech

- Gained practical exposure to cybersecurity concepts and security testing methodologies.
- Worked with cybersecurity tools and practical security exercises.
- Performed vulnerability-analysis and web-security activities.
- Used Burp Suite for security testing and proof-of-concept development.
- Prepared technical documentation and security reports.
- Developed practical understanding of penetration testing, vulnerability assessment, and secure application practices.

PROJECTS

Spidy AI Trading Platform | AI/ML | Python | PyTorch | XGBoost | MT5 | Shoonya | Streamlit

- Designed an AI-assisted algorithmic trading platform integrating MT5 Bridge and broker APIs.
- Developed a modular StrategyManager architecture supporting multiple trading strategies.
- Explored PyTorch and XGBoost for market prediction, signal generation, and strategy intelligence.
- Designed a Streamlit-based dashboard for monitoring trading strategies, signals, and performance.
- Planned online-learning capabilities to allow models and strategies to adapt to changing market conditions.
- Integrated automated trading workflows with risk-management and strategy-selection concepts.

Supervisory Analytics Tool for SOC Assessment (SAT-SA) | Cybersecurity | SOC | Security Analytics | SIH Project

- Developed a solution concept for assessing and improving Security Operations Center maturity and cyber resilience.
- Designed workflows for SOC monitoring, incident analysis, threat detection, response, and security assessment.
- Proposed centralized analytics to evaluate SOC performance and identify operational weaknesses.
- Designed dashboards and security metrics to support SOC analysts, administrators, and decision-makers.
- Worked on the project as a solution aligned with Smart India Hackathon Problem Statement 26157.

AI-Driven Emergency Response System – LifeLink | AI | IoT | Blockchain | Emergency Response

- Designed an AI-powered emergency response platform for intelligent accident and medical emergency management.

- Proposed Edge AI-based injury assessment to prioritize emergency response.
- Designed ambulance-routing logic and Green Corridor traffic coordination.
- Integrated blockchain concepts for secure legal authorization and emergency data integrity.
- Designed a system architecture connecting patients, hospitals, emergency services, and authorities.

Secure Smart Vault System | Raspberry Pi | IoT Security | NFC | OTP | Sensors | GPIO

- Developed a hardware-based secure vault prototype combining NFC authentication, OTP verification, sensors, and electronic locking.
- Integrated Raspberry Pi with NFC modules, accelerometer/gyroscope sensors, magnetic door sensors, relay modules, and solenoid locks.
- Designed a multi-stage authentication workflow: NFC → OTP → Verification → Relay Activation → Solenoid Unlock → Sensor Monitoring → Auto Relock.
- Implemented security logic for failed authentication attempts with buzzer and red LED alerts.
- Developed a touch-friendly security interface for an 800×480 display.
- Worked with Raspberry Pi GPIO, I²C devices, power regulation, and hardware-software integration.

BrowserShield – Secure Browser Protection Concept | Cybersecurity | Browser Security | Python | Authentication

- Designed a browser security solution focused on protecting sensitive browser sessions and authentication tokens.
- Proposed mechanisms for cookie/cache cleanup and removal of active login sessions.
- Designed access-control concepts using password and biometric authentication.
- Explored automation for browser security management using Python.
- Focused on reducing risks associated with unauthorized access to authenticated web sessions.

GPU-Accelerated Hardware-Locked Password Recovery Utility | Cybersecurity | Python/Java | GPU Computing | NVIDIA RTX

- Designed a forensic/recovery-oriented password recovery utility with GPU acceleration.
- Proposed hardware-locked operation using a USB device as an authorization mechanism.
- Designed separate Forensic, Research, and Recovery operating modes.
- Explored NVIDIA GPU monitoring and acceleration using an RTX 3050.
- Designed a PySide6-based multi-page desktop interface for security-focused workflows.

TECHNICAL SKILLS

Programming: Python, Java, C, SQL, JavaScript, HTML, CSS

Cybersecurity: SOC Operations, Threat Detection, Incident Analysis, Vulnerability Assessment, Web Security, Security Monitoring, Network Security, Secure Software Development

Security Tools: Burp Suite, Nmap, Bettercap, Wazuh, Kali Linux, Wireshark

AI / Machine Learning: PyTorch, XGBoost, Machine Learning, AI-based Threat Detection, Computer Vision, Predictive Analytics

Web & Backend: Next.js, Node.js, FastAPI, Streamlit, REST APIs

Hardware / IoT: Raspberry Pi, NFC/PN532, RC522, MPU6050, RTC, GPIO, Relay Modules, Solenoid Locks, Sensors

Databases: PostgreSQL, SQL, Supabase

Development Tools: Git, GitHub, VS Code, Linux, VirtualBox

CYBERSECURITY EXPERIENCE & PRACTICAL WORK

SOC Incident Investigation: Hands-on analysis of simulated security incidents involving phishing emails and malicious URLs; suspicious domains and impersonation attacks; proxy/network security events; endpoint security alerts; malicious USB devices; malware detection and quarantine; user and source-IP investigation; threat intelligence correlation; incident escalation and response.

Vulnerability Assessment & Web Security: Performed web application security testing using Burp Suite. Worked with vulnerability identification, request/response analysis, and proof-of-concept documentation. Practiced reconnaissance and service enumeration using Nmap. Worked with network-security testing environments using Kali Linux and

Bettercap. Prepared structured vulnerability reports containing vulnerability description, CWE classification, CVSS assessment, affected endpoint, reproduction steps, proof of concept, impact, evidence, remediation.

ACADEMIC & TECHNICAL EXPERIENCE

Developed applications and prototypes using Python, Java, C, SQL, JavaScript, HTML and CSS. Worked with data structures including linked lists, trees, graphs, BFS, DFS, AVL trees, and sorting/searching concepts. Studied secure software engineering principles and database security. Worked with Linux environments, Kali Linux, VirtualBox, SSH, and development environments. Built hardware prototypes involving Raspberry Pi, GPIO, sensors, NFC, relays, and electronic locks. Designed project architectures, database schemas, security workflows, UI wireframes, and technical documentation.

KEY AREAS OF INTEREST

Security Operations Center (SOC); Cyber Threat Intelligence; Incident Response; Vulnerability Assessment; Penetration Testing; Web Application Security; AI for Cybersecurity; Machine Learning; Secure Software Engineering; IoT Security; Digital Forensics; Security Automation; Cloud & Application Security

SOFT SKILLS

Problem Solving; Analytical Thinking; Technical Research; Team Collaboration; Project Development; Presentation & Documentation; Troubleshooting; Rapid Learning

CAREER OBJECTIVE

To build a career in Cybersecurity, SOC Operations, AI-driven Security, or Secure Software Engineering, where I can apply practical security knowledge, programming skills, and project experience to identify threats, develop secure systems, automate security operations, and contribute to real-world cybersecurity solutions.